Welcome to the second costume design guild. Todays topic is materials, textiles and natural dying

Transparency

My master research

- Running these guilds as part of my master research
- Needs to be recorded for documentation

Block B:

- Test runs of “workshop” set-up
- Practise teaching
- **Fashion content**
- In-person only

Block C:

- “Real” runs of my research
- Data gathering
- **Costume content (for games)**
- In-person but also streamed

My master research is about the gap in knowledge of foundational fashion/costume knowledge in the current game design education
I am running these guilds as part of my master research and it needs to be recorded for documentation

Block B is about fashion content because I need to testrun this workshop setup, which is also why its in-person only so I can practise teaching this way
And block C will be the real runs of my research with costume content, designed for game contexts. These guilds will also be streamed so you don’t have to miss something if you cant join in person

PRE-SESSION QUESTIONS

Red:
- No
- No experience
- Unfamiliar
- Not confident

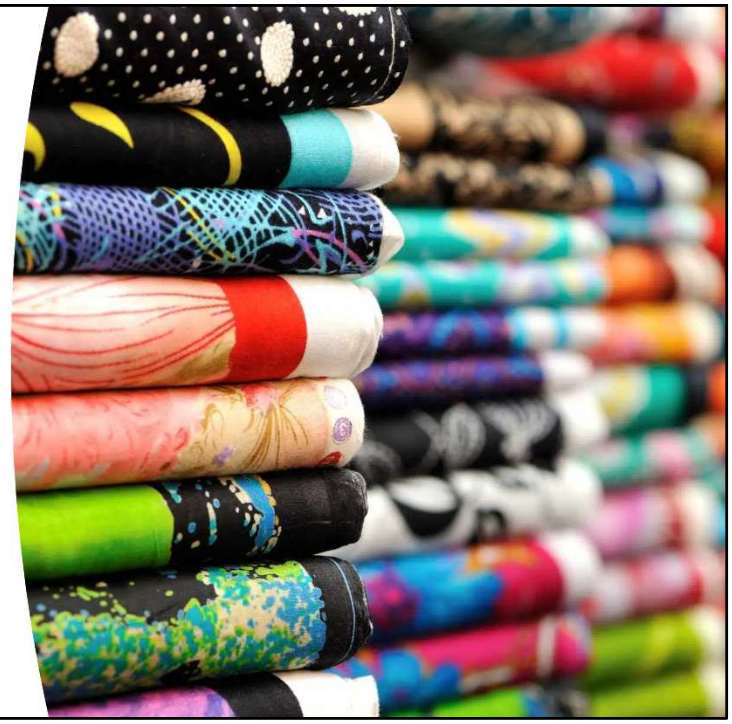
Yellow:
- Maybe
- Some experience
- Somewhat familiar
- Moderately confident

Green:
- Yes
- Experienced
- Very familiar
- Very confident

Lets start with some questions which will be part of my data collection for my master thesis.
At the end of this presentation we will do this again
- Can you tell the difference between a knitted fabric and a woven fabric?
- Could you label what textiles/materials your clothing is made from right now?
- Do you know what natural fibers look like before they become fabric? (e.g., what does a cotton plant look like?)
- Have you heard that you can dye fabric with food items like onions or avocados?
- How confident do you feel understanding how fabric choice affects costume design? (scale)

Agenda

- Material choice = storytelling
- Why this matters for game design
- Material pipeline
- Fiber types
- Fabric construction
- Activity 1: textile exploration
- Natural dyes
- Activity 2: weaving exploration
- Conclusion



This is the agenda for this guild
I will be uploading this presentation to teams so you can always find it again if it goes too fast for notes
By I suggest taking notes of things you think are especially useful for your own work so you can search online for more information later

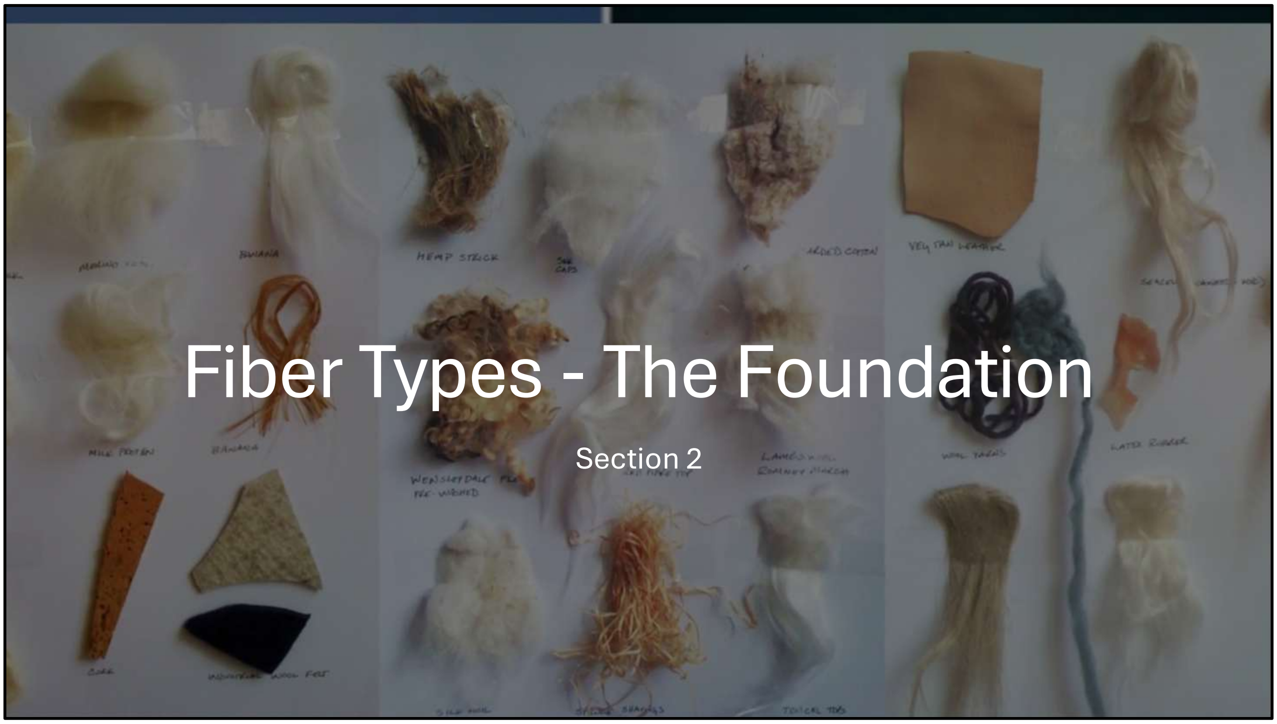
I want to start with an introduction about why materials matter

The Material Problem

Material choice = visual storytelling through texture, drape, weight, movement

So you're designing characters with different costumes - but do you know WHY a heavy wool cloak behaves differently than flowy satin robes?
•[appear] Material choice = visual storytelling through texture, drape, weight, movement
•Today: Understanding what fabric actually IS and how it affects character design

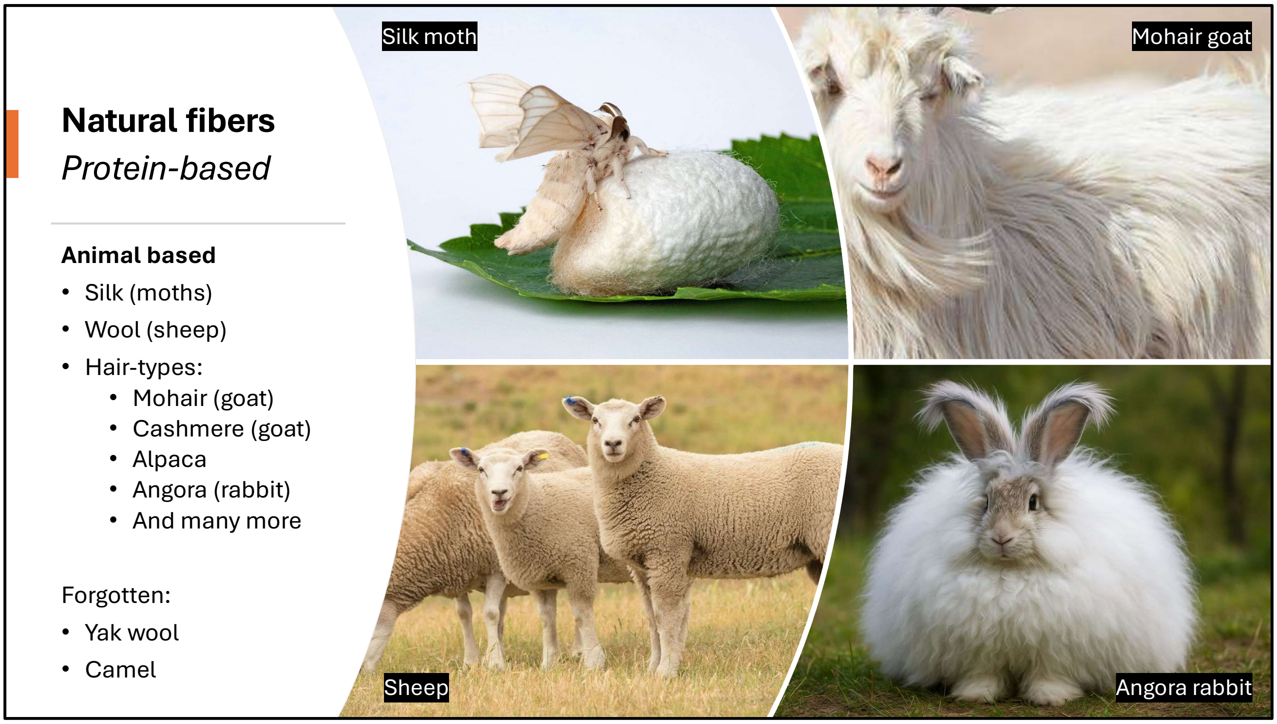
• Same garment design in different materials = a completely different silhouette/mood
• Heavy armor padding (thick quilted fabric) vs flowing mage robes (lightweight silk)
• Weathered traveler (rough linen, natural dyes) vs wealthy merchant (fine weaves, vibrant synthetic dyes)
Material choice tells your character's story before they say a word



Let’s talk about fiber types

Natural Fibers are cellulose based
Plant-based: Cotton (fluffy bolls), linen (flax stalks), hemp, ramie

Properties: breathable, absorb moisture, take dye well, natural texture



So if your game world has hairy creatures, consider making it a fiber source for garments

Natural Fibers :
Animal-based: Wool (sheep fleece), silk (cocoons), cashmere, mohair
There are also some forgotten wool types like yak wool and camel
So if your game has some strange fantastical animal, consider making it a wool-giving animal and create garments with it.

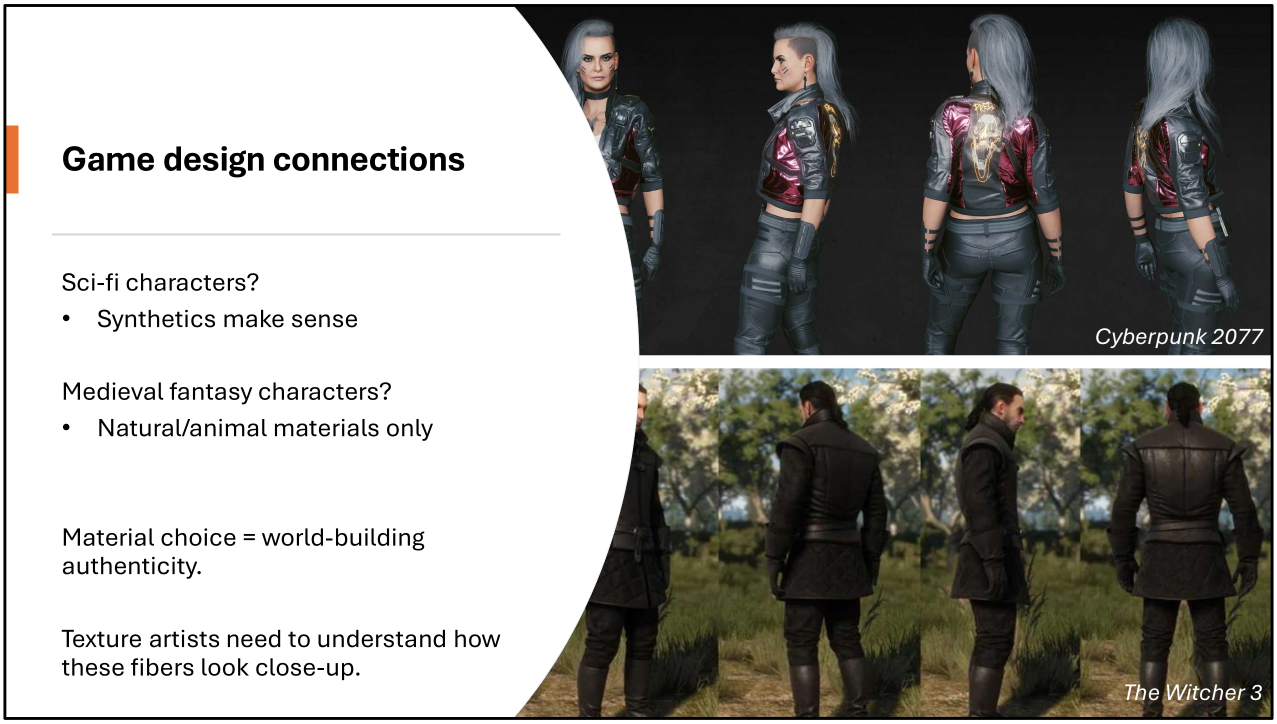
Properties: breathable, absorb moisture, take dye well, natural texture

Synthetic fibers
Synthetic-based

Fossil-fuel based
- Polyester
- Nylon
- Acrylic
- Spandex/elastane

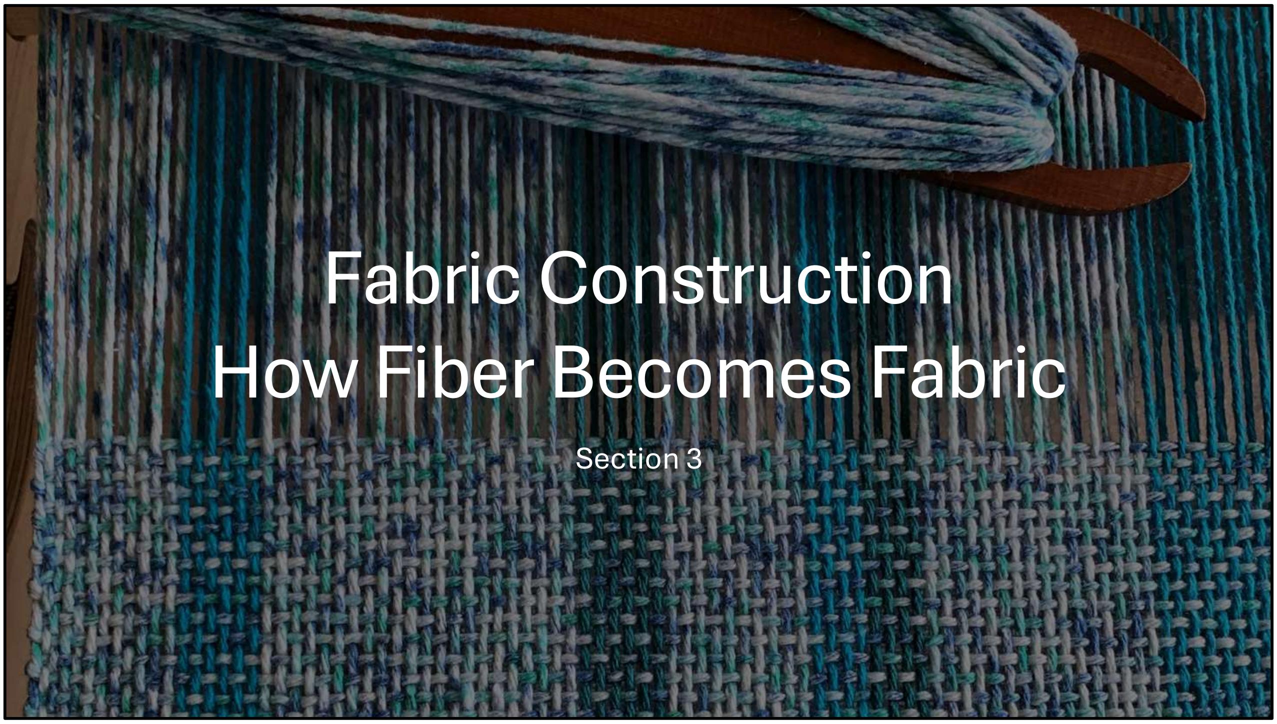
Synthetic Fibers
Polyester, nylon, acrylic, spandex/elastane
Quite bad for the environment

Properties: stretchy, water-resistant, colour-fast, smooth texture, doesn't breathe
Invented in 20th century - if your character is pre-industrial, they can't have these!

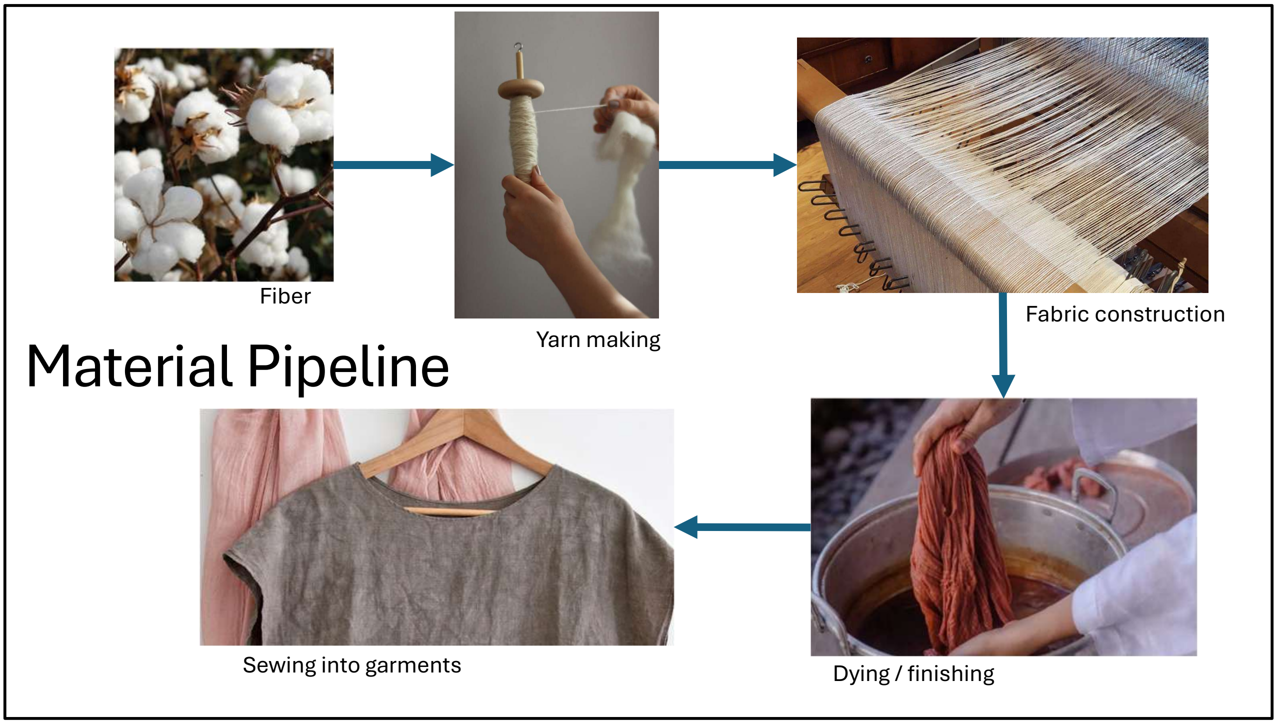


Game Design Connection

- Medieval fantasy characters? Natural fibers only (cotton, linen, wool, silk)
- Sci-fi characters? Synthetics make sense (technical fabrics, performance wear)
- Material choice = world-building authenticity
- Texture artists need to understand how these fibers look close-up

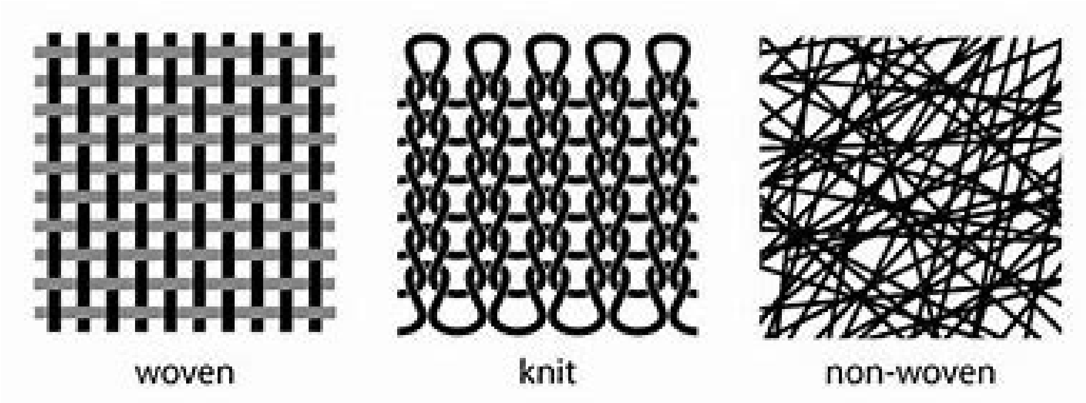


The next topic is fabric construction and I will talk about how fibers become fabric



Material Pipeline Overview
Fiber gets harvested and through processing methods necessary, it gets turned into something spinable
→ Yarn gets spun
→ Fabric Construction, happens when yarn is constructred into fabrics, I will talk about different constructions later
→ Dyeing/Finishing of the yarn, or of the fabric
→ Final Costume is sewn
•"We're covering the full journey today - from plant to dyed fabric"

The 3 construction methods



Weaving
(interlacing threads at right angles)

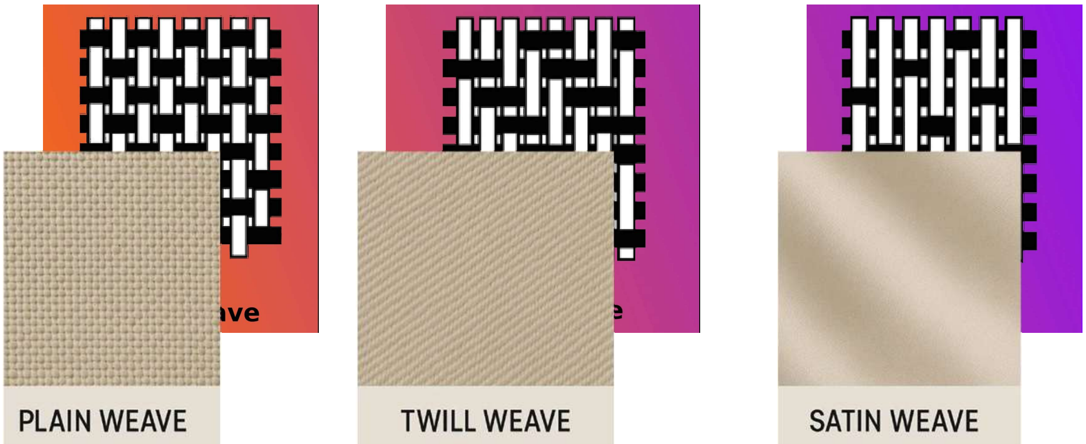
Knitting
(looping yarn together)

Non-wovens
(felting, bonding, laminating)

The Three Construction Methods

1. **Weaving** (interlacing threads at right angles)
2. **Knitting** (looping yarn together)
3. **Non-wovens** (felting, bonding, laminating)

Different weave structures



- **Plain weave:** Over-under-over-under - canvas, muslin
- **Twill weave:** Diagonal pattern - denim, gabardine
- **Satin weave:** Smooth, lustrous surface - satin, sateen

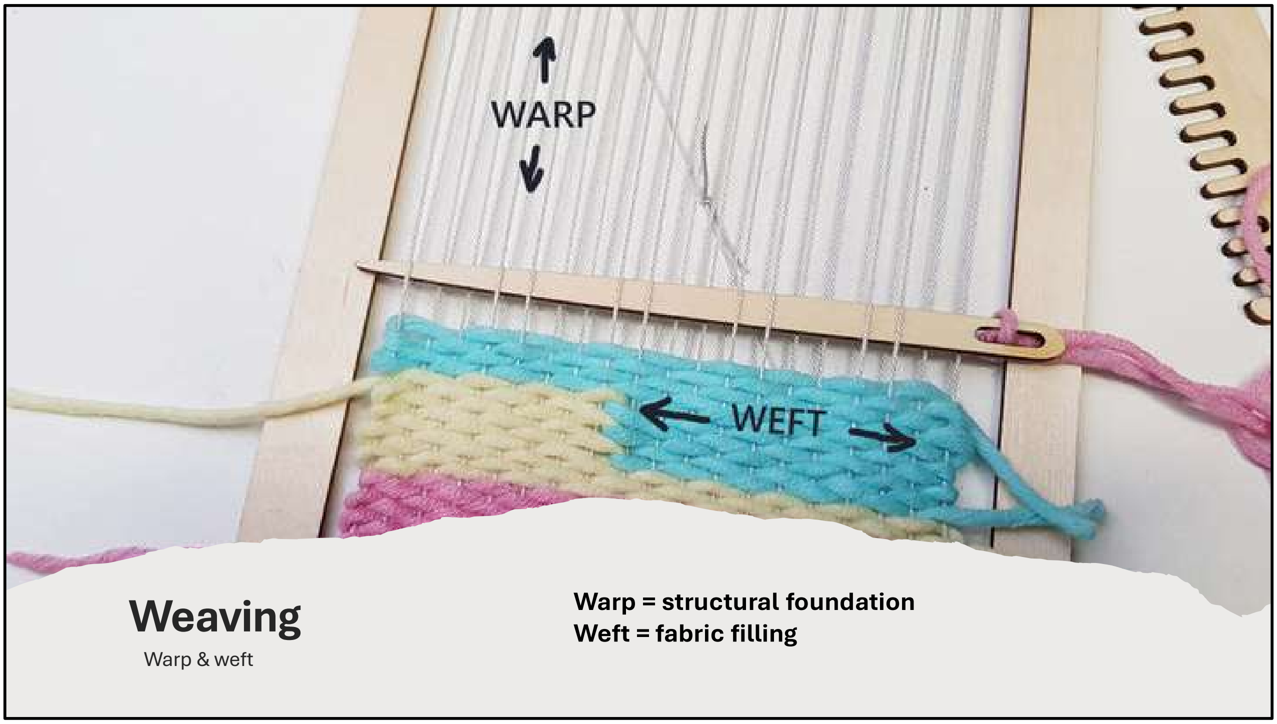
Slide 9: Weaving Deep-Dive Visuals of different weave structures:

•**Plain weave:** Over-under-over-under (strongest, most stable) - denim, canvas, muslin

•**Twill weave:** Diagonal pattern (durable, drapes well) - denim, chino, gabardine

•**Satin weave:** Smooth, lustrous surface (elegant but delicate) - satin, sateen

•Close-up photos of each so you can see what the construction image turned into real fabric



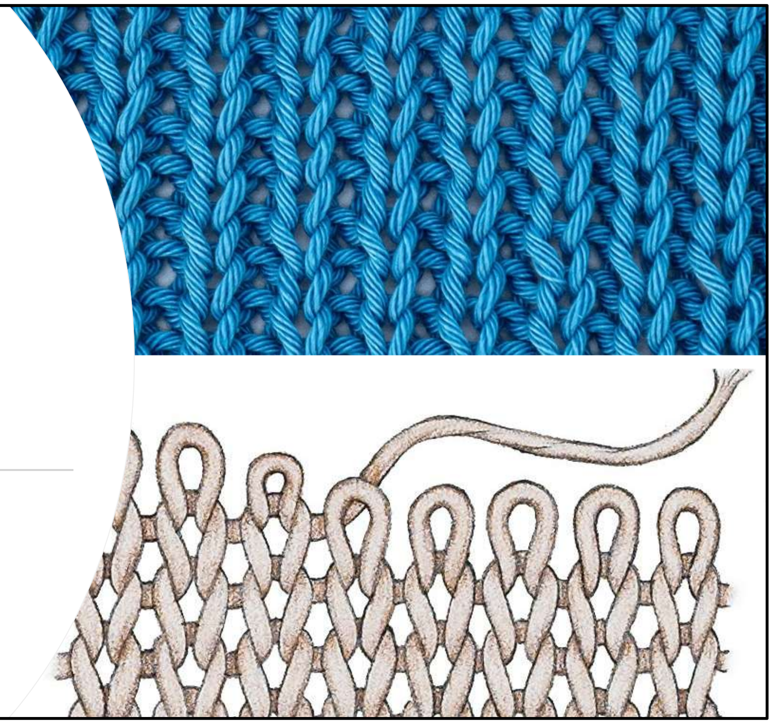
Warp is the structural foundation of the fabric - vertically
And weft is the horizontal thread that fills the fabrics, creates patterns and dictate texture and colour

Knitting Overview

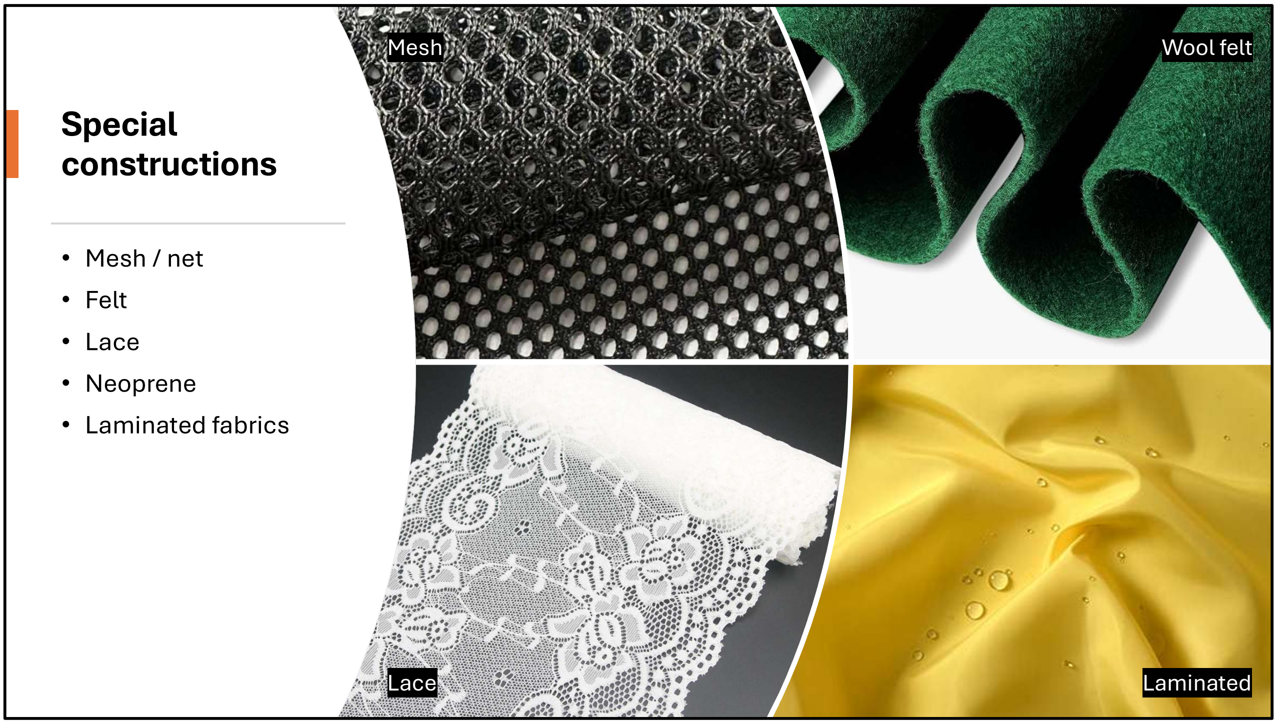
Key property:

Stretch and recovery

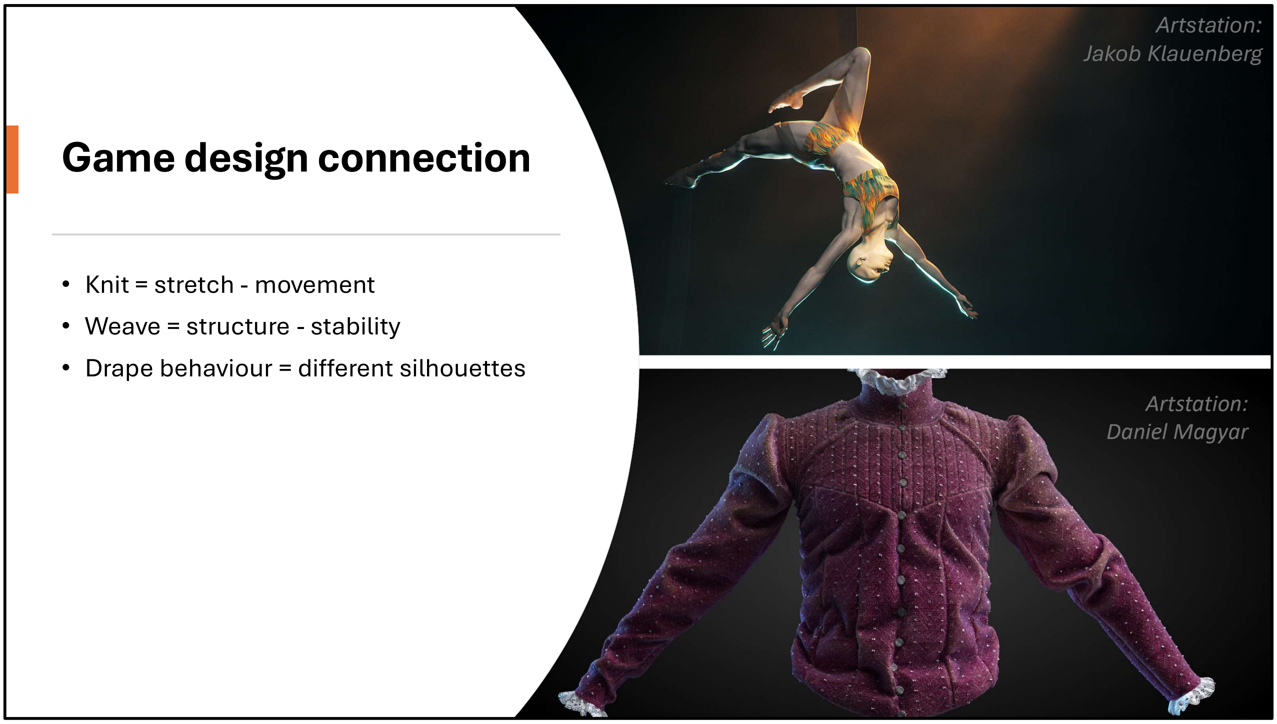
Knits move with the body



Slide 10: Knitting Overview
Knitting by hand is done with just one yarn, which loops through the previous row of loops
•**Weft knitting** (rows, stretchy) - t-shirts, sweaters, socks
•**Warp knitting** (vertical, more stable) - athletic wear, lingerie
•Key property: **stretch and recovery** - knits move with the body
•Show visual of how pulling one thread can unravel a knit vs a weave



Slide 11: Specialty Constructions Quick showcase:
•**Mesh/net:** Open structure (athletic jerseys, veils)
•**Felt:** Non-woven, matted fibers (hats, craft projects)
•**Lace:** Decorative openwork (wedding dresses, historical costumes)
•**Neoprene:** Bonded layers (wetsuits, modern performance wear) - fabric + foam + fabric sandwich
•**Laminated fabrics:** Fabric bonded to other materials (raincoats, technical outerwear)



Slide 12: Game Design Connection Examples with visuals:

•**Knit = stretch:** If your character does acrobatics, their costume needs knit components or they couldn't move realistically

•**Weave = structure:** Medieval armor, tailored coats, fantasy cloaks - these need woven fabric stability

•**Drape behavior:** Tight weaves (stiff, hold shape) vs loose weaves (fluid, flow) = different silhouettes

•Reference image: Same cape design in stiff canvas vs flowing silk - completely different character vibe

Time for an activity
I’ve prepared a bunch of fabrics for you guys to touch and compare and see if you can recognise some of them already

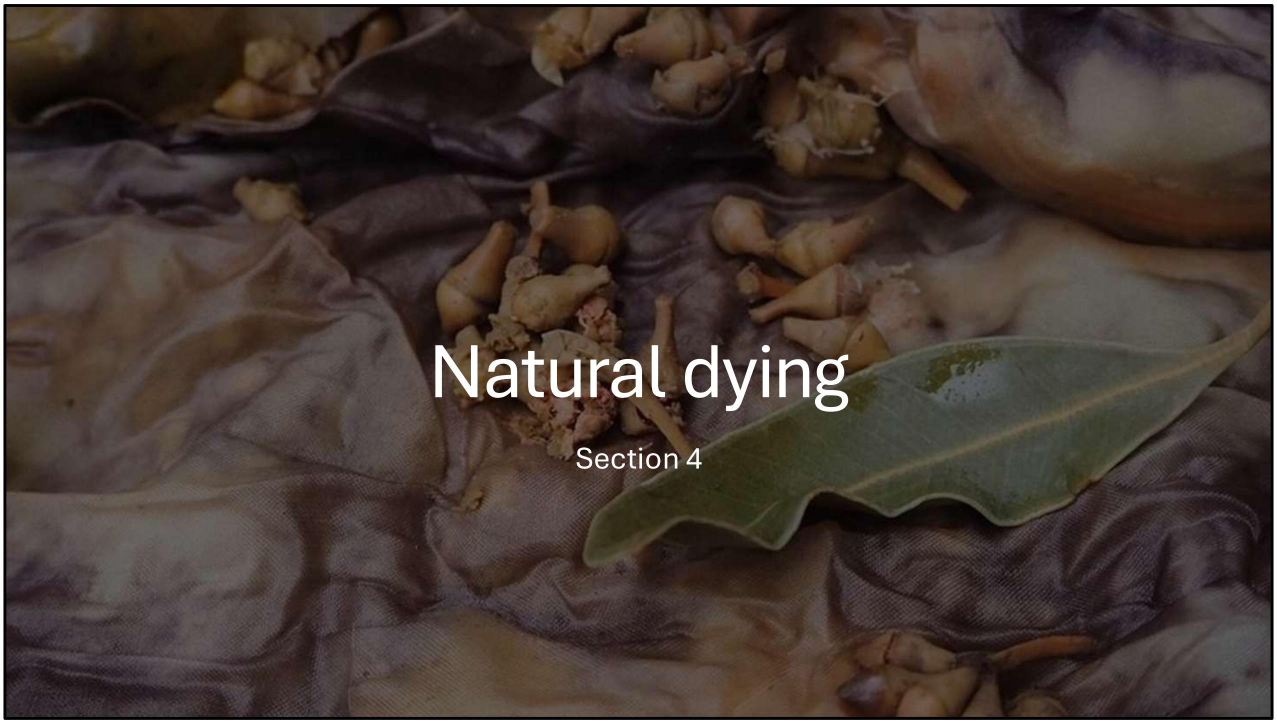
Activity timeline

- **Exploration (5 min):**

Guided prompts:

- Can you identify which are knits and which are woven?
- Can you identify what material it's made of?
- Compare a very fluid fabric to a stiff one
  How would these behave differently on a character?
- Look at the twill weave pattern
  Can you see the diagonal?

- **Debrief (2 min):**
- What clues helped?
- What was confusing?
- Any fabric particularly interesting?



My final topic is natural dying

What is natural dying?

Before synthetic dyes (invented 1856), all colour came from plants, insects, minerals

Colours:

- Blue: woad plant
- Red: madder roots
- Purple: snails
- Scarlet: kermes scale insect

Historical context: expensive dyes = wealth indicator

What is Natural Dyeing?
•Before synthetic dyes (invented 1856), all color came from plants, insects, minerals
•Historical context: expensive dyes = wealth indicator **(Tyrian purple for royalty, indigo for luxury)**
•Your fantasy/historical characters? Their costume colors tell their social class

•10,000 to 20,000 snails to make 1 gram of Tyrian purple dye

You can also dye fabrics with food items you can find in your kitchen
As seen here, onions, avocado, spinach, lemon, beets, red cabbage and blueberries

Fiber matters

Silk dyes most vibrantly – Protein amino acids in silk create multiple bonding sites

Cotton dyes well – Less bonding sites in cellulosic fibers, needs more mordant

Tencil dyes okay – Manufacturing process creates too smooth fiber, resisting dye bonding

Synthetic does not dye - fabrics would stain (surface) but not bond like protein/cellulose fibers

As you might have seen during the textile exploration, I dyed the same 7 fabrics with different dyestuffs

Natural fibers (cotton, linen, silk, wool): Take dye beautifully, vibrant color

•**Synthetic fibers (polyester, nylon):** Barely dye at all, stay pale/gray

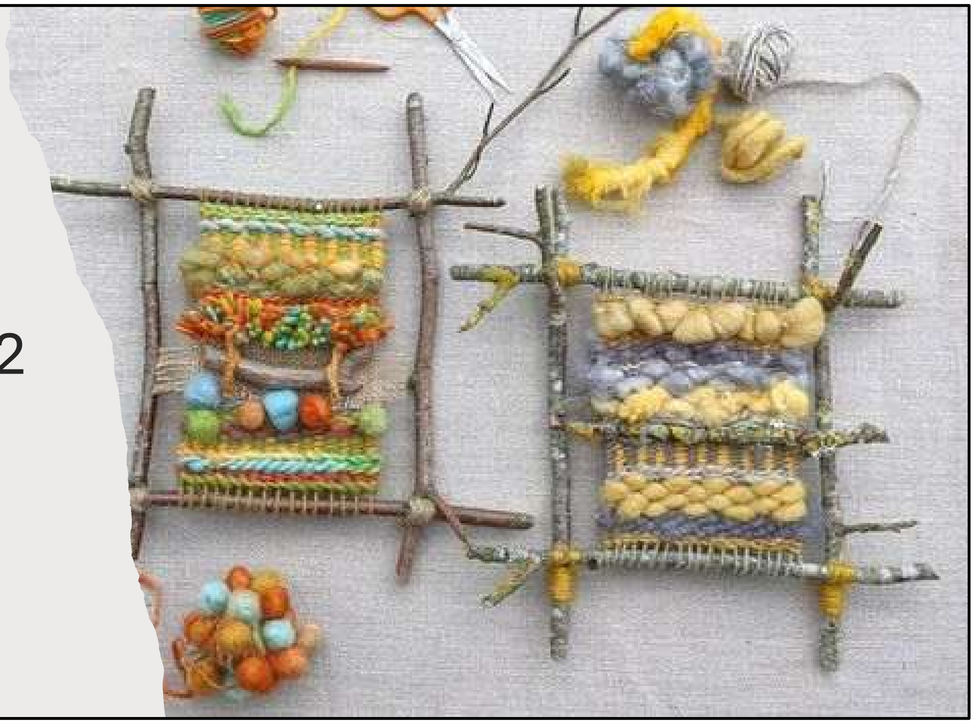
•Show same dye bath on different fibers: "See how silk is bright but polyester is nearly unchanged?“

•Synthetics struggle to take in natural dyes and only stain (which leaves when in sun or washed) because the synthetic fibers do not bond to dye

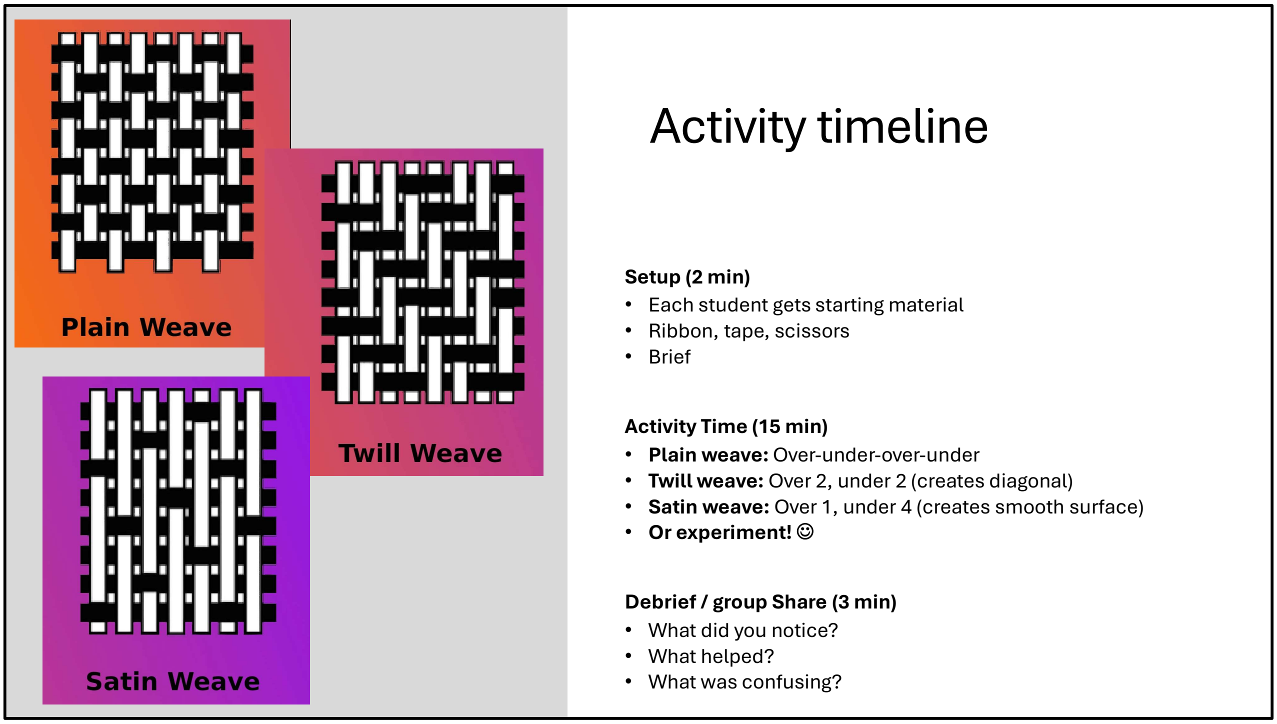
•Mordanting (helps dye bind to fiber, alum, metals)
•Dye bath (simmering plant material) straining it
•Fiber absorption (time + heat) sometimes cooking for several hours and then leaving it overnight
•Rinsing + setting (alum or vinegar)
•Drying, ready for use
•"The process is slow and unpredictable - part of its beauty and historical value"

Activity 2

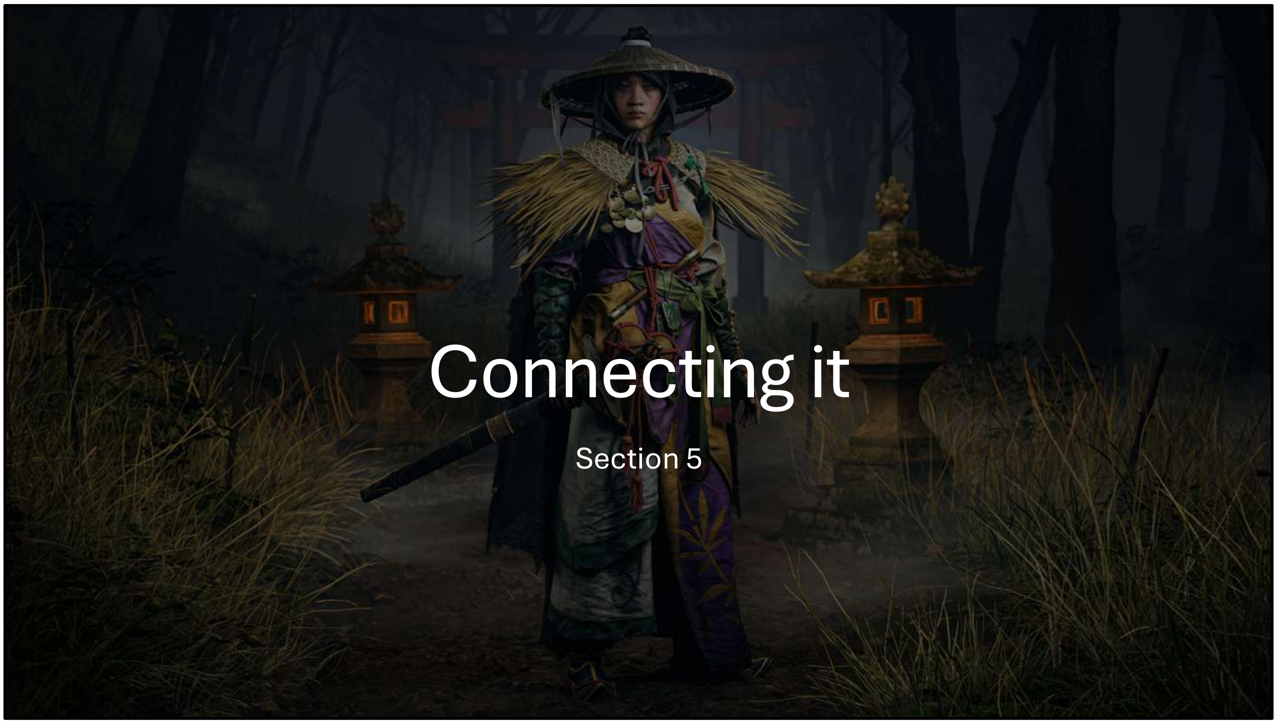
Weaving exploration



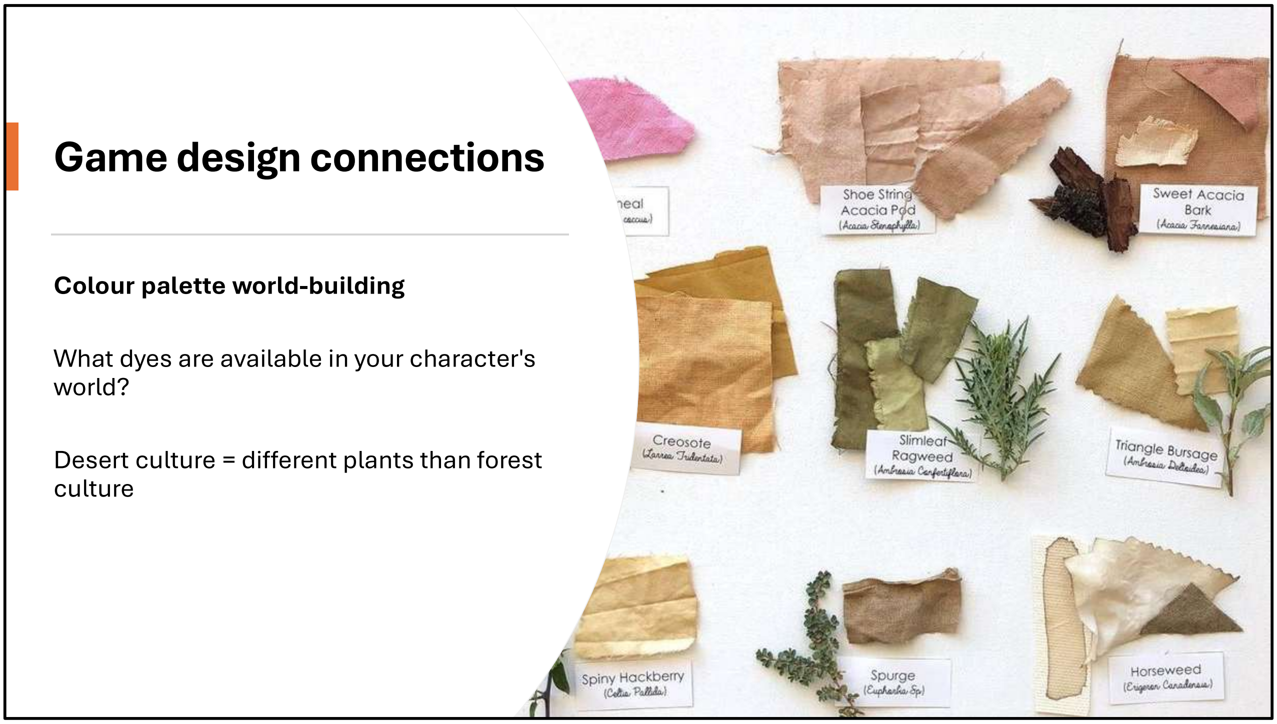
Time for another activity, we are going to do a weaving exploration


Plain Weave
Twill Weave
Satin Weave
Activity timeline
Setup (2 min)
• Each student gets starting material
• Ribbon, tape, scissors
• Brief
Activity Time (15 min)
• Plain weave: Over-under-over-under
• Twill weave: Over 2, under 2 (creates diagonal)
• Satin weave: Over 1, under 4 (creates smooth surface)
• Or experiment! ☺
Debrief / group Share (3 min)
• What did you notice?
• What helped?
• What was confusing?

See slide



Finally, I want to quickly connect it to game design again



Game Design Connection
Color palette world-building: What dyes are available in your character's world? Desert culture = different plants than forest culture

Game design connections

Weathering/fading

- Fabrics weather/distress with usage
- Natural dyes fade realistically with sun/water exposure

How does this show character age/travel?

Game Design Connection
Weathering/fading: Natural dyes fade realistically with sun/water exposure - how does this show character age/travel?

Game design connections

Class indicators

- Bright, saturated colours = expensive dyes = wealthy character
- Dull earth tones = common dyes = peasant class

Historical accuracy: If your game is set in 1400s, your colour palette should reflect available natural dyes

Game Design Connection

Class indicators: Bright, saturated colors = expensive dyes = wealthy character. Dull earth tones = common dyes = peasant class

Historical accuracy: If your game is set in 1400s, your color palette should reflect available natural dyes

Recap using viking as example

From Fiber to Costume Visual summary showing the full journey:

•Cotton plant → spun yarn → woven fabric → dyed with onions → finished costume

•"Every costume has this journey - understanding it makes you a better character designer"

Quick recap:

And how you can use this in your work

- **Material choice** affects character movement, silhouette and story
- **Fiber type** determines dye possibilities and historical accuracy
- **Weave/knit/other** impacts drape and flexibility
- **Natural dyes** = limited historical colour palettes, location-locked = world-building tool

What This Means for Your Work

•Material choice affects character movement, silhouette, and story
•Fiber type determines dye possibilities and historical accuracy
•Weave/knit structure impacts drape and flexibility
•Natural dyes = limited historical color palettes = world-building tool

POST-SESSION QUESTIONS

Red:
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Yellow:
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- Can you tell the difference between a knitted fabric and a woven fabric?
- Could you label what textiles/materials your clothing is made from right now?
- Do you know what natural fibers look like before they become fabric?
- Have you heard that you can dye fabric with food items?
- How confident do you feel understanding how fabric choice affects costume design? (scale)

Plus possibly:
- Do you feel this knowledge would help you design more believable character costumes?
- Would you be interested in learning more about textiles?

Thanks for joining, hope to see you next week!

Same room same time! ☺

Topics of next week:

- Digital fashion
  + Live demo CLO/MD

Interesting sources

- Forgotten fibers [link]
- Medieval dyes [link]
- Viking dyes [link]
- Dyes in Early Northern Europe [link]
- Wikipedia list of natural dyes [link]